

Computer Networking Basics Activity

1. IP Address:

Purpose: Shows the IP address and network configuration of your system.

Output: Shows Network interface, IP address, subnets masks.

2. Ping:

Purpose: It tests the connectivity of your computer and another device.

Output: Displays the reply packets(both transmitted and received), response time and packet loss data.

3. Traceroute:

Purpose: It shows the path taken by the data packets to reach the destination.

Output: List every router between your computer and the destination with the time taken.

4. Nslookup:

Purpose: It resolves the domain name into the corresponding IP address.

Output: Displays the DNS server and IP Address of the requested domain.

5. Netstat:

Purpose: Displays the list of active network and the listening ports.

Output: Shows the protocol(tcp/udp), local addresses, port numbers, and the status whether its listening or not.

6. Network Diagram:



7. How a request travels from your computer to a website?

- ➔ 1. Whenever a user enters a website address(e.g. google.com) in a browser, the browser first sends a request to a DNS server to fetch the website's IP address.
2. Request is then forwarded from user's computer to the router, which sends it over to the internet.
3. The request reaches the destination server, which processes it and sends the requested webpage back through the internet & router to the user's computer and is displayed in browser.

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purvaa@Purva:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet 10.255.255.254/32 brd 10.255.255.254 scope global lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host proto kernel_lo
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:15:5d:19:2d:b9 brd ff:ff:ff:ff:ff:ff
    altname enx00155d192db9
    inet 172.23.58.136/20 brd 172.23.63.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::215:5dff:fe19:2db9/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 56:1b:87:14:a7:0e brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever
4: flannel.1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1450 qdisc noqueue state UNKNOWN group default
    link/ether 8e:aa:9d:b1:c9:e2 brd ff:ff:ff:ff:ff:ff
    inet 10.42.0.8/32 scope global flannel.1
        valid_lft forever preferred_lft forever
    inet6 fe80::8caa:9dff:feb1:c9e2/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
5: cnio0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1450 qdisc noqueue state UP group default qlen 1000
    link/ether 72:4c:26:0f:b5:b8 brd ff:ff:ff:ff:ff:ff
    inet 10.42.0.1/24 brd 10.42.0.255 scope global cnio0
        valid_lft forever preferred_lft forever
    inet6 fe80::704c:26ff:fe0f:b5b8/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
6: veth663fa940i2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1450 qdisc noqueue master cnio0 state UP group default
    link/ether ae:5d:5d:cf:82:a5 brd ff:ff:ff:ff:ff:ff link-netns cn1-0202576d-a883-c222-511a-92e15f81688b
    inet6 fe80::dcd9:a7ff:fe83:c602/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
7: veth6acc77140i2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1450 qdisc noqueue master cnio0 state UP group default
    link/ether 4a:51:fc:46:96:b8 brd ff:ff:ff:ff:ff:ff link-netns cn1-ada281aa-1e43-cab9-2216-b61107a170d3
    inet6 fe80::4851:fcff:fe46:96b8/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
8: vethc3d09980i2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1450 qdisc noqueue master cnio0 state UP group default
    link/ether 4e:f7:3d:94:cd:67 brd ff:ff:ff:ff:ff:ff link-netns cn1-29e01a82-6877-1db3-bfef-c3fa16f00f01
    inet6 fe80::4cf7:3dff:fe94:cd67/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever

purvaa@Purva:~$ ping google.com
PING google.com (142.251.222.142) 56(84) bytes of data:
 64 bytes from pnm3aa-az-in-f14.1e100.net (142.251.222.142): icmp_seq=1 ttl=117 time=56.0 ms
 64 bytes from pnm3aa-az-in-f14.1e100.net (142.251.222.142): icmp_seq=2 ttl=117 time=71.0 ms
 64 bytes from pnm3aa-az-in-f14.1e100.net (142.251.222.142): icmp_seq=3 ttl=117 time=68.8 ms
 64 bytes from pnm3aa-az-in-f14.1e100.net (142.251.222.142): icmp_seq=4 ttl=117 time=72.1 ms
 64 bytes from pnm3aa-az-in-f14.1e100.net (142.251.222.142): icmp_seq=5 ttl=117 time=125 ms
 64 bytes from pnm3aa-az-in-f14.1e100.net (142.251.222.142): icmp_seq=6 ttl=117 time=50.5 ms
 64 bytes from pnm3aa-az-in-f14.1e100.net (142.251.222.142): icmp_seq=7 ttl=117 time=57.6 ms
 64 bytes from pnm3aa-az-in-f14.1e100.net (142.251.222.142): icmp_seq=8 ttl=117 time=55.3 ms
 64 bytes from pnm3aa-az-in-f14.1e100.net (142.251.222.142): icmp_seq=9 ttl=117 time=60.2 ms
 64 bytes from pnm3aa-az-in-f14.1e100.net (142.251.222.142): icmp_seq=10 ttl=117 time=105 ms
 64 bytes from pnm3aa-az-in-f14.1e100.net (142.251.222.142): icmp_seq=11 ttl=117 time=55.7 ms
 64 bytes from pnm3aa-az-in-f14.1e100.net (142.251.222.142): icmp_seq=12 ttl=117 time=114 ms
 64 bytes from pnm3aa-az-in-f14.1e100.net (142.251.222.142): icmp_seq=13 ttl=117 time=82.5 ms
 64 bytes from pnm3aa-az-in-f14.1e100.net (142.251.222.142): icmp_seq=14 ttl=117 time=60.3 ms
 64 bytes from pnm3aa-az-in-f14.1e100.net (142.251.222.142): icmp_seq=15 ttl=117 time=57.1 ms
 64 bytes from pnm3aa-az-in-f14.1e100.net (142.251.222.142): icmp_seq=16 ttl=117 time=55.6 ms
^C
--- google.com ping statistics ---
16 packets transmitted, 16 received, 0% packet loss, time 15174ms
rtt min/avg/max/mdev = 55.341/72.166/125.013/21.971 ms
purvaa@Purva:~$ traceroute google.com
traceroute to google.com (142.251.222.142), 30 hops max, 60 byte packets
 1 Purva.mshome.net (172.23.48.1) 0.773 ms 0.696 ms 0.656 ms
 2 Unit (192.168.1.1) 9.667 ms 9.645 ms 9.598 ms
 3 10.240.164.13 (10.240.164.13) 36.911 ms 47.240 ms 47.300 ms
 4 172.28.3.203 (172.28.3.203) 40.764 ms 172.28.3.202 (172.28.3.202) 48.231 ms 46.868 ms
 5 dsl-mp-dynamic-061.129.246.61.airtelbroadband.in (61.246.129.61) 46.830 ms 46.816 ms dsl-mp-dynamic-057.129.246.61.airtelbroadband.in (61.246.129.57) 46.766 ms
 6 116.119.161.135 (116.119.161.135) 46.744 ms 182.79.141.205 (182.79.141.205) 43.292 ms 116.119.73.209 (116.119.73.209) 43.271 ms
 7 * * *
 8 * * *
 9 142.251.64.10 (142.251.64.10) 55.322 ms 142.250.228.50 (142.250.228.50) 49.260 ms 55.256 ms
10 192.178.110.206 (192.178.110.206) 55.211 ms 72.14.232.78 (72.14.232.78) 55.112 ms 55.079 ms
11 192.178.254.237 (192.178.254.237) 55.051 ms 192.178.254.31 (192.178.254.31) 57.059 ms 192.178.254.221 (192.178.254.221) 55.750 ms
12 216.229.49.46 (216.229.49.46) 78.319 ms 216.229.49.130 (216.229.49.130) 71.755 ms 172.253.64.226 (172.253.64.226) 68.178 ms
13 142.251.230.71 (142.251.230.71) 77.510 ms 172.253.71.3 (172.253.71.3) 77.420 ms 142.251.230.91 (142.251.230.91) 77.310 ms
14 209.85.253.85 (209.85.253.85) 55.952 ms 63.465 ms 74.880 ms
15 pnm3aa-az-in-f14.1e100.net (142.251.222.142) 69.046 ms 69.833 ms 74.843 ms
purvaa@Purva:~$ nslookup google.com
Server:
 10.255.255.254
Address:
 10.255.255.254#53
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purvaa@Purva: -  
10 192.178.110.206 (192.178.110.206) 55.211 ms 72.14.232.78 (72.14.232.78) 55.112 ms 55.079 ms  
11 192.178.254.237 (192.178.254.237) 55.051 ms 192.178.254.31 (192.178.254.31) 57.059 ms 192.178.254.221 (192.178.254.221) 55.750 ms  
12 216.239.49.46 (216.239.49.46) 78.319 ms 216.239.49.130 (216.239.49.130) 71.755 ms 172.253.64.226 (172.253.64.226) 68.178 ms  
13 142.251.230.71 (142.251.230.71) 77.510 ms 172.253.71.3 (172.253.71.3) 77.420 ms 142.251.230.91 (142.251.230.91) 77.310 ms  
14 209.85.253.85 (209.85.253.85) 55.952 ms 63.465 ms 74.880 ms  
15 pnaaaa-az-in-f14.1e100.net (142.251.222.142) 69.046 ms 69.033 ms 74.843 ms  
purvaa@Purva:~$ nslookup google.com  
Server: 10.255.255.254  
Address: 10.255.255.254#53  
  
Non-authoritative answer:  
Name: google.com  
Address: 142.251.222.142  
Name: google.com  
Address: 2404:6800:4007:831::200e  
  
purvaa@Purva:~$ netstat -tuln  
Active Internet connections (only servers)  
Proto Recv-Q Send-Q Local Address Foreign Address State  
tcp 0 0 0 127.0.0.1:10010 0.0.0.0:* LISTEN  
tcp 0 0 0 127.0.0.54:53 0.0.0.0:* LISTEN  
tcp 0 0 0 127.0.0.1:6444 0.0.0.0:* LISTEN  
tcp 0 0 0 127.0.0.1:59421 0.0.0.0:* LISTEN  
tcp 0 0 0 127.0.0.1:10257 0.0.0.0:* LISTEN  
tcp 0 0 0 127.0.0.1:10256 0.0.0.0:* LISTEN  
tcp 0 0 0 127.0.0.1:10259 0.0.0.0:* LISTEN  
tcp 0 0 0 127.0.0.1:10258 0.0.0.0:* LISTEN  
tcp 0 0 0 127.0.0.1:10249 0.0.0.0:* LISTEN  
tcp 0 0 0 127.0.0.1:10248 0.0.0.0:* LISTEN  
tcp 0 0 0 10.255.255.254:53 0.0.0.0:* LISTEN  
tcp 0 0 0 127.0.0.53:53 0.0.0.0:* LISTEN  
tcp6 0 0 0 :::80 :::* LISTEN  
tcp6 0 0 0 :::6443 :::* LISTEN  
tcp6 0 0 0 :::10250 :::* LISTEN  
tcp6 0 0 0 :::8080 :::* LISTEN  
udp 0 0 0 127.0.0.54:53 0.0.0.0:*  
udp 0 0 0 127.0.0.53:53 0.0.0.0:*  
udp 0 0 0 10.255.255.254:53 0.0.0.0:*  
udp 0 0 0 0.0.0.0:8472 0.0.0.0:*  
udp 0 0 0 127.0.0.1:323 0.0.0.0:*  
udp 0 0 0 127.0.0.1:323 0.0.0.0:*  
udp6 0 0 0 :::1:323 :::*  
udp6 0 0 0 :::1:323 :::*
```

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Date: 24 July 2026

Batch: B1